RUI ZHOU

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EDUCATION

IOWA STATE UNIVERSITY — Ames, IA
Ph.D., Materials Science and Engineering — Expected 2027

WASHINGTON UNIVERSITY IN ST. LOUIS — St. Louis, MO
M.S., Materials Science and Engineering — December 2019

CHANGCHUN UNIVERSITY OF SCIENCE AND TECHNOLOGY — Jilin, China
B.S., Materials Science and Engineering — June 2018

RESEARCH EXPERIENCE

IOWA STATE UNIVERSITY — Ames, IA
Graduate Research Assistant — January 2023 - Present

- Apply first-principles calculations and machine-learning force fields to investigate the structure, phase stability, and transport behavior of solid-state electrolyte and structural materials.
- Perform large-scale molecular dynamics simulations to study phase transitions, ion diffusion, and structure–transport relationships in energy materials.
- Develop and use computational workflows for atomistic simulation, data analysis, and materials-property prediction.

WASHINGTON UNIVERSITY IN ST. LOUIS — St. Louis, MO
Graduate Research Assistant — August 2018 – December 2019

- Studied the influence of surface properties on thermal and mass-transfer behavior during water evaporation using molecular dynamics simulations.
- Investigated nanoscale evaporation mechanisms and interfacial transport behavior through atomistic simulation.
- Optimized micropillar-array geometries to improve evaporation and cooling performance.

PROFESSIONAL EXPERIENCE

Sunwoda Electric Vehicle Battery Co., ltd — Shenzhen, China
Simulation Engineer — December 2020 - December 2022

- Developed simulation-based methods to support lithium-ion battery development and manufacturing, with a focus on safety, stability, and energy density.
- Developed tools to fit model parameters to experimental data, reducing the cost and time required for model calibration.
- Applied computational modeling and data analysis to support battery R&D, manufacturing optimization, and failure-mechanism analysis.

PUBLICATIONS

R Zhou, K Luo, Q An. *Atomic-Scale Mechanisms of Interphase Formation at Lithium–Glassy Sulfide Electrolyte Interfaces*. Journal of Materials Chemistry A, 2026

R Zhou, K Luo, Q An. *Anion Topology and Cation–Anion Coupling Govern Phase Transitions and Ionic Transport in Closo-Hydroborate Solid Electrolytes*. ACS Applied Energy Materials, 2026

R Zhou, K Luo, Q An. *Data-Driven Atomistic Modeling of Crystalline and Glassy Solid-State Electrolytes*. Chemical Communications 61 17254-17270, 2025

K Luo, **R Zhou**, Q An. *Mechanistic insights into photoplasticity of CdTe*. International Journal of Mechanical Sciences 305 110765, 2025

J Zheng, …, **R Zhou**, … H Wang. *dpdata: A Scalable Python Toolkit for Atomistic Machine Learning Data Sets*. Journal of Chemical Information and Modeling 65 (21) 11497-11504, 2025

R Zhou, K Luo, L Fei, Q An. *Unraveling Superionic Conductivity in Na2B12H12: Molecular Dynamics Study of Phase Transition, Anion Reorientation, and Ionic Conductivity via Machine Learning Force Field*. ACS Electrochemistry 1 (2) 143-152, 2025

K Luo, **R Zhou**, Q An. *Unraveling Photoplasticity in ZnS: Enhanced Peierls Stress under Photoexcitation using Machine Learning Potentials*. ACS Materials Letters 7(1) 46-51, 2024

K Luo, **R Zhou**, SW Martin, Q An. *Flexible doorway controlled Na+ ionic diffusion in NaPSO glassy electrolytes from machine-learning force field simulations*. Journal of Materials Chemistry A 12 (48) 33518-33525, 2024

R Zhou, K Luo, SW Martin, Q An. *Insights into Lithium Sulfide Glass Electrolyte Structures and Ionic Conductivity via Machine Learning Force Field Simulations*. ACS Applied Materials & Interfaces 16 (15) 18874-18887, 2024

J Liu, K Luo, **R Zhou**, Q An. *Understanding the Role of 1/2 ⟨110⟩ Dislocations in Deformation Mechanisms of Single-Crystal High-Entropy Carbide Ceramics from Machine Learning Force Field Simulations*. ACS Applied Engineering Materials 2 (7) 1857-1865, 2024

H Wang, C Yuan, **R Zhou**, Q Duan, Y Li. *Self-sacrifice template formation of nitrogen-doped porous carbon microtubes towards high performance anode materials in lithium ion batteries*. Chemical Engineering Journal 316 1004-1010, 2017

J Guo, J Liu, H Dai, **R Zhou**, T Wang, C Zhang, S Ding, H Wang. *Nitrogen doped carbon nanofiber derived from polypyrrole functionalized polyacrylonitrile for applications in lithium-ion batteries and oxygen reduction reaction*. Journal of colloid and interface science 507154-161, 2017

PATENTS

"Ternary material and preparation method thereof, lithium ion battery and power utilization equipment"
(Granted) CN114744187B. Date of Application: June 2022

"Method, system, equipment and storage medium for optimizing particle size of material particles"
(Granted) CN114169233A. Date of Application: Nov 2021

"Simulation optimization design method, system, equipment and medium for battery material mixing proportion"
(Granted) CN115579085A. Date of Application: Oct 2022

"Parameter identification method, system, device and medium for secondary battery physical model"
(Granted) CN115392123A. Date of Application: Aug 2022